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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

I / II Semester B. E. Regular / Supplementary Examinations Feb -2024

Common to AS / CH / IM / ME

CLASSICAL PHYSICS FOR ENGINEERS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer SIX full questions from Part B. In Part B question number 2 and 11 are compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8 & 9 and 10.
3. Handbook of Physics is allowed.

PART-A

1	1.1	Define Poisson's ratio.	01
	1.2	What is the strain of an incompressible body?	01
	1.3	In spite of having shock absorber, why drivers go slowly over bumps?	01
	1.4	What is the damping term in mechanical oscillators?	01
	1.5	What is the solid to gas phase transition process called?	01
	1.6	What is irreversible process in Thermodynamics?	01
	1.7	Give an example of capillary motion.	01
	1.8	What is viscosity of a fluid?	01
	1.9	What are photoelectrons?	01
	1.10	Define Tensile strength of a material.	01

PART-B

2	a	With a neat labeled diagram, setup and solve the differential equation governing the dynamics of an under-damped oscillator. Represent the solution graphically.	10
	b	A $2D$ square plate of mass $1kg$ and length $10cm$ is suspended by a wire which passes through its face center and perpendicular to its plane. Find the time period of angular oscillation for small displacement if the torque required per unit twist of the wire is $4.5 \times 10^{-3}\ N-m/radian$.	04
3	a	With a neat diagram, derive an expression for the couple per unit twist of a solid shaft of radius R and length L. What would be the difference if we take a hollow shaft instead, with inner radius R_1 and outer radius R_2?	10
	b	Calculate the sharing modulus of a body which has a bulk modulus of $8.0 \times 10^{10} N/m^2$ and Poisson's ratio of 0.25.	04
		OR	
4	a	Define briefly any 2 types of stress and strain. Show that a shearing stress is equal to a tensile stress and a compressive stress.	10
	b	A load of $5kg$ produces an extension of $1mm$ in a wire of $5m$ in length and $1mm$ in diameter. Calculate the Young's modulus of the wire.	04

5	a	Derive an expression for the work done in a adiabatic process, after deriving the adiabatic gas law equation. State 2 limitations of the 1st law of thermodynamics.	10
	b	When the valve of a completely evacuated bottle is opened, $0.2m^3$ of the atmospheric air at a pressure of $101.325kPa$ rushes in. Calculate the work done by air.	04
		OR	
6	a	With a diagram explain Joule's experiment demonstrating the equivalence of heat and work. Show how it leads to the concept of internal energy, hence explain the first law of thermodynamics for a cyclic process.	10
	b	A stationary mass of gas is compressed isobarically $(P = 200kPa)$ without friction from $0.3m^3$ to $0.15m^3$. Calculate the change in internal energy if there is a transfer of $38.7kJ$ of heat from the gas during the process.	04
7	a	Diagrammatically explain the difference between stream line, streak line and path line. Derive the continuity equation in Integral form.	10
	b	A stream of water emerging from a faucet of area $1.2cm^2$, falls through a vertical distance of $85mm$ under gravity whereby its area become $0.12cm^2$. Calculate the rate at which water flows from the faucet.	04
		OR	
8	a	Discuss briefly the construction, working principle and procedure of an ultrasonic interferometer.	10
	b	A $1mm$ diameter glass tube is inserted into water at 20^0C in a cup. Determine the capillary rise of water in the tube. Given the surface tension of water at 20^0C is $0.073N/m$. Given density of water is $1000kg/m^3$. Assume contact angle for water is 10^0.	04
9	a	Describe in details the optical and $I-V$ characteristics techniques of a material.	10
	b	What is yield strength of material? What method is used to test it?	04
		OR	
10	a	Explain the principle, construction and working of an atomic force microscope with neat diagram. Mention two of its applications.	10
	b	Discuss the basic working principle and use of a X ray diffractometer.	04
		LAB COMPONENT	
11	a	In an experiment on torsional pendulum, the following data was obtained	

Configuration	Time for 10 oscillations
Circular disc ($R = 7.5\ cm$), Mass = $1.1\ kg$ Wire passing through centre of gravity And perpendicular to its plane	i) $39.1\ s$ ii) $40.2\ s$ iii) $38.9\ s$
Rectangular disc (length $15cm$, breadth $7cm$) wire is attached perpendicular to both length and breadth mass = $1.2\ kg$	i) $24.5\ s$ ii) $24.2\ s$ iii) $24.1\ s$

b	Calculate the rigidity modulus of the wire, given the length of the wire is $20cm$ and radius of the wire is $0.3mm$	10
	By constructing a ray diagram, explain the apparatus required and procedure involved to find the wavelength of a laser beam by diffraction technique and calculate the wavelength of a laser beam from the data given below. Given the grating constant is $5.08 \times 10^{-5}m$ and the distance between grating and screen is $100cm$.	10

Diffraction order (n)	*Distance* $2x_n$
1	2.6
2	5.4
3	8.2
4	10.8
5	13.8
6	16.4

Academic year 2023-2024
(TEST I FOR FIRST SEM B.E. PROGRAMS)

DEPARTMENT OF PHYSICS

Date	November 20, 2023	Maximum Marks	50
Course Code	PHY211BI	Duration	90min
Sem	First Semester (AS, ME, IEM, CH & SPARK)	Test-I	
CLASSICAL PHYSICS FOR ENGINERS			

Physical constants: $h=6.625\times10^{-34}$Js, $m_e=9.1\times10^{-31}$kg and Boltzmann constant $k=1.38\times10^{-23}$ J/K, Mass of neutron or proton$=1.67\times10^{-27}$ kg. $\varepsilon_0=8.85\times10^{-12}$ Fm^{-1}.

Sl. No.	Questions	M	BT	CO
1a.	Setup second order differential equation for damped harmonic oscillation and obtain the solution for displacement. Discuss under damping condition.	07	2	CO2
1b	A body executes Simple harmonic oscillation such that its velocity at the mean position is 1 m/s and acceleration at one of the extremity is 1.57 m/s^2. Find the time period of vibration.	03	4	CO3
2 a)	Obtain an expression for effective spring constant of two identical mass less springs connected in i) series and ii) parallel. Comment on time periods of oscillation.	07	3	CO2
2b	Calculate the peak amplitude of vibration of a system whose natural frequency is 1000 Hz when it oscillates in a resistive medium for which the value of damping/unit mass is 0.008 rad/s under the action of an external periodic force/unit mass of amplitude 5 N/kg.	03	4	CO3
3a	Show that a simple shear is equal to the sum of equal elongation strain and compression strain mutually perpendicular to each other.	06	3	CO1
3b	A steel wire with 30 cm length for which Young's modulus is $1.6\text{x}10^{11}$ N/m^2 has a 0.75 mm diameter. If a load of 1.45 kg is attached to one end, how much is the extension produced?. If Poisson's ratio is 0.27, what is the lateral compression produced?	04	4	CO3
4a	Derive an expression for couple per unit twist of a solid cylinder having radius 'R' and length 'L'.	08	2	CO2
4b	Find the rigidity modulus of gold if its Young's modulus is $7.0\text{ x}10^{10}$ N/m^2 and Poisson' ratio is 0.425.	02	3	CO3
5a	Assuming bending moment, obtain an expression for depression produced for a load 'W' at the free end of single cantilever having circular cross section.	06	4	CO2
5b	Calculate the couples per unit twist for a 27 cm long wire with 0.8 mm diameter and rigidity modulus $6.8\text{x}10^{10}$ N/m^2. If the same wire is designed to perform torsion oscillation, compute the time period of torsion pendulum if its free end is connected to a circular disc of circumference 47 cm and mass 1200 g with the axis of rotation perpendicular to plane of the circular disc and passing through its centre.	04	2	CO3

Course Outcome	CO1	CO2	CO3
Max. Marks	06	28	16

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

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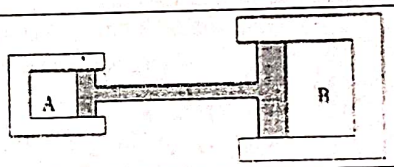
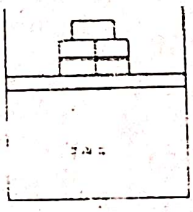
DEPARTMENT OF PHYSICS

Course: Classical Physics for Engineers	CIE: II First semester 2023-24	Maximum marks: 50 Time: 90 Minutes
Course code: PHY211BI	Physics Cycle: Mechanical Stream	Date: 27/12/2023

Instructions to candidates:

i. ***Answer all the questions***

ii. *Physical constants:*

Q.No	PART B – Test	M	BT	CO
1a.	Obtain the relationship between viscosity and drag force. Use this principle to explain the construction and working of concentric cylinder rotational viscometer with a neat figure.	7	L2	2
1b.	If $P_A = P_B$, in which direction will the piston move? Explain using equations. 	3	L4	3
2a.	Define zeroth law of thermodynamics? Discuss the principle and working of constant volume gas thermometers.	7	L2	1
2b.	A chamber contains 0.04 m^3 of a random gas with an initial pressure of 200 N/m^3 supplied by the weights on the piston. Keeping this pressure constant, the chamber is heated until the volume of the gas is 0.1 m^3. While heating is going on, weights are removed from the piston in such a way that the relation between pressure and volume is given by the expression PV = constant. Calculate the work done by the system. 	3	L1	3
3a.	Derive the adiabatic equation of state and work done in adiabatic expansion of a gas.	7	L3	2
3b.	The two arms of a U tube have diameters 2mm and 1 mm. The tube is partly filled with water and is held with its arms vertical. Find the difference in the levels of water in the two limbs if the surface tension of water is 70×10^{-3} Nm^{-1}	3	L1	3
4a.	Define surface tension and derive the excess pressure inside a droplet and soap bubble using a neat figure.	7	L3	2
4b.	A liquid is compressed isothermally inside a chamber. Obtain an expression for the total amount of work required if the compression process is quasistatic and is given by the equation $\ln (V / V_0) = -A (p - p_0)$, where A, V_0 and p_0 are constants.	3	L3	3
5a.	Discuss the construction and working of ultrasonic interferometer with a neat figure	7	L1	1
5b.	Draw the model graphs for an isochoric, isobaric and isothermal process for an ideal gas.	3	L2	1

COs	CO 1	CO 2	CO 3
Marks	17	21	12

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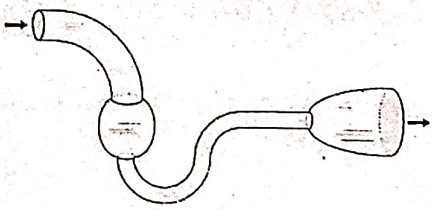
Academic year 2023-2024

(TEST III FOR FIRST SEM B.E. PROGRAMS)

DEPARTMENT OF PHYSICS

Date	January 22, 2024	Maximum Marks	50
Course Code	PY211IB	Duration	90min
Sem	First Semester (AS, ME, IEM, CH & SPARK)	Improvement Test	
CLASSICAL PHYSICS FOR ENGINERS			

Physical constants: $h=6.625\times10^{-34}$Js, $m_e=9.1\times10^{-31}$kg and Boltzmann constant $k=1.38\times10^{-23}$ J/K, Mass of neutron or proton$=1.67\times10^{-27}$ kg. $\varepsilon_0=8.85\text{x}10^{-12}$ Fm^{-1}.

Sl. No.	Questions	M	BT	CO
1a.	Setup second order differential equation for damped harmonic oscillation and obtain the solution for displacement. Discuss critical damping condition.	07	2	CO1
1b	A car has a spring system that supports the in-built mass 1000 kg. When a person with a weight 980 N sits at the centre of gravity, the spring system sinks by 2.8 cm. When the car hits a bump, it starts oscillation vertically. Find the period and frequency of oscillation.	03	4	CO3
2 a)	Obtain an expression for time period of a body executing angular simple harmonic motion.	07	3	CO2
2b	A sonometer wire under tension is plucked and left free for vibrations. Find its frequency of vibrations if the midpoint on the string attains a maximum velocity of 1.57 m/s when its amplitude of oscillation is 5 mm. Treat the vibration as simple harmonic. (Neglect the damping effect)	03	4	CO3
3a	Explain Laminar and Turbulent flows of fluid with one example each.	06	3	CO1
3b	 In the above figure, water enters the pipe at 6 m/s and flows out at 1.5 m/s. Both ends of the pipe have circular area of cross section and the entry has 1 cm radius. Find the radius of the pipe at the exit end.	04	4	CO3
4a	With a neat labeled diagram, explain the construction and working of ellipsometry technique employed in optical characterization of thin films.	06	2	CO2
4b	Give a short note on magnetic particle testing.	04	1	CO2
5a	What is UTM? Explain construction and working of UTM.	08	2	CO2
5b	A circular steel bar of 20 mm diameter carries a tensile load of 30KN. Find the tensile stress in the bar.	02	3	CO3

Course Outcome	CO1	CO2	CO3
Max. Marks	10	25	15

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks